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3) Do you think that option is a type of a future?

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4) When can the call option and put option be exercised?

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13.8 SWAPS

In finance, a swap refers to a financial contract between two parties to exchange cash flows or financial instruments. In this contract, the parties agree to swap one type of cash flow or financial instrument for another based on predetermined terms. These agreements encompass both spot and forward transactions within a customized framework. Corporations, banks, and individual investors now utilize swaps for intricate financing solutions, reducing borrowing costs, and gaining better control over interest rate and foreign currency risks. Swaps serve diverse purposes, ranging from hedging to speculation, but their core objective is to alter the nature of an asset or liability without the need for liquidation. For instance, an investor deriving returns from equity investments can transform those returns into less risky fixed-income cash flows without liquidating the equities. Similarly, a corporation with floating-rate debt can convert it into a fixed-rate obligation without retiring and reissuing debt.

Various features of Swaps are as follows:

1) **Customizability:** Swaps are highly customizable agreements tailored to meet the specific needs of the parties involved.

2) **Counterparty Risk:** Swaps involve counterparty risk since the agreement is dependent on both parties fulfilling their obligations.

3) **Over-the-Counter (OTC) Trading**: Swaps are typically traded OTC, which means they are not traded on standardized exchanges but rather through a network of dealers.

4) **Types of Swaps:** The most common types of swaps include interest rate swaps, currency swaps, and commodity swaps. Each type has different underlying assets and cash flow structures.

5) **Fixed and Floating Rates**: Swaps often involve the exchange of fixed interest rates for floating rates (or vice versa), helping parties manage interest rate exposure.

6) **Maturity:** Swaps have specified maturity dates, which can range from a few months to several years, depending on the agreement.

7) **Notional Principal**: The notional principal amount is the total value on which the exchanged cash flows are calculated. This principal itself is usually not exchanged.

8) **Cash Flow Exchange**: Parties exchange cash flows based on the agreed-upon terms, which may occur periodically (e.g., quarterly, semi-annually).

9) **Purpose:** Swaps are used for hedging risks, managing exposure to fluctuations in interest rates or currencies, and speculating on future price movements.

10) **Documentation**: Swaps are documented in a swap agreement, which includes terms such as the notional amount, payment dates, calculation methods, and conditions for default.

Let us understand various forms of swaps. Common swaps are:

a) **Interest rate swap –** Under this type of swap, one party agrees to pay fixed interest on a notional amount and receive floating interest on the same notional amount. Generally, the floating interest rate is calculated as LIBOR (London Inter Bank Offered Rate) + Some base points. In India, MIBOR (Mumbai Inter Bank Offered Rate) is also popular in place of LIBOR.

Example 13.5: Suppose there are two companies A and B. They enter an agreement with each other in which company A will pay Company B a fixed interest at 7% p.a. on a notional amount of ₹10,00,000 and it will receive floating interest i.e., LIBOR + 1%. The agreement is valid for 3 years. If the LIBOR data is given below, then the cash flows under the interest rate swap can be calculated as follows:

Years	**LIBOR**
0 to 1 year	8%
1 to 2 years	6%
2 to 3 years	10%

Step 1: Floating Interest

Years	**Floating interest rate = LIBOR + 1%**	**Floating interest (in ₹)**
1	9%	90,000
2	7%	70,000
3	11%	1,10,000

Step 2: Fixed Interest

Years	Floating interest rate = LIBOR + 1%	Floating interest (in ₹)
1	10%	100,000
2	10%	100,000
3	10%	100,000

Step 3: Exchange of net cash flows

Years	Net cash flows (in ₹)	Explanation
1	(-) 20,000	Company A will pay ₹ 100,000 and receive ₹ 80,000 from B, therefore, A can pay a net ₹ 20,000.
2	(-) 30,000	Company A will pay ₹ 1,00,000 and receive ₹ 70,000 from B, therefore, A can pay a net ₹ 30,000.
3	10,000	Company A will pay ₹ 1,00,000 and receive ₹ 1,10,000 from B, therefore, A can receive a net ₹ 10,000.

b) **Credit Default Swaps (CDS)** – A credit default swap (CDS) is a financial contract between two parties that provides insurance against the risk of default on a specific debt instrument, such as a bond or loan. The buyer of the CDS makes payments to the seller of the CDS in exchange for a guarantee that in the event of a default by the borrower, the seller will pay the buyer the face value of the debt instrument.

Example 13.6: Suppose Company A issues a bond worth ₹10 million with a maturity of 5 years. Investors who buy the bond are essentially lending money to Company A, and in return, Company A agrees to make regular interest payments and repay the principal at maturity. However, there is a risk that Company A may default on its debt obligations, which would mean that investors would not receive their interest payments or the principal at maturity. To mitigate this risk, an investor can purchase a CDS from a financial institution, such as a bank or hedge fund. The investor pays a premium to the institution to protect against the risk of default by Company A. If Company A defaults, the institution must pay the investor the face value of the bond, which is ₹10 million in this case. The investor effectively transfers the risk of default from themselves to the institution, which is compensated through the premium paid by the investor.

Check Your Progress 3:

1) State the key features of SWAP.

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2) Discuss the potential of SWAPS in India.

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13.9 LET US SUM UP

Derivatives: Financial contracts that derive their value from an underlying asset or security, such as stocks, bonds, or commodities constitute the derivatives. They have various pros and cons. Their biggest advantage is **hedging** against the risk and the biggest disadvantage is default risk. The various types of derivatives include Futures, Forwards, Options, Swaps etc.

Futures are Standardized contracts that obligate the buyer to purchase and the seller to sell a specific asset at a predetermined price and date in the future. Similar to futures contracts, **Forwards** are not standardized and are usually customized to meet the specific needs of the buyer and seller. **Options** are Contracts that give the buyer the right, but not the obligation, to buy or sell an underlying asset at a predetermined price and date in the future. Agreements between two parties to exchange cash flows based on different financial instruments or indexes, such as interest rates or currencies are **Swaps**. There are various swap agreements, but interest rate swaps and credit default swaps are the two most popular ones. Under an interest rate swap agreement, two parties agree to exchange fixed interest for floating interest, while a credit default swap (CDS) is a financial contract that allows one party to transfer the credit risk of a particular asset or security to another party. In essence, it is a type of insurance policy that protects the buyer of the CDS from the risk of default by the issuer of the underlying security. Derivatives trading began in India in 2000, and the National Stock Exchange (NSE) and Bombay Stock Exchange (BSE) are the two exchanges in India where derivatives are traded.

Derivatives in India are traded on two main exchanges: the National Stock Exchange (NSE) and the Bombay Stock Exchange (BSE). The Securities and Exchange Board of India (SEBI) regulates the derivatives market. Commodity derivatives are traded on the Multi Commodity Exchange (MCX) and the National Commodity and Derivatives Exchange (NCDEX), while currency derivatives are traded on the Metropolitan Stock Exchange (MSE).

13.10 KEY WORDS

Derivative : A financial asset that derives its value from another asset called an underlying asset. There are various types of derivatives. It includes futures, forwards, options, commodity derivatives, etc.

Futures : Futures contracts are financial instruments that allow parties to buy or sell an asset, such as commodities, currencies, or financial instruments, at a predetermined price and date in the future. The price of the underlying asset is fixed at the time the futures contract is created, and the transaction is legally binding.

Forwards : Forwards are similar to futures, but they are not traded on exchanges. Instead, they are customized contracts between two parties to buy or sell an underlying asset at a predetermined price and time in the future.

Option : An option is a right to buy or sell an underlying at a strike or exercise price at the expiry of the option.

Swap : In finance, a swap is a derivative contract that allows two parties to exchange financial instruments or cash flows with each other based on agreed-upon terms. Interest rate swaps and credit default swaps are two very popular swaps.

13.11 SOME USEFUL BOOKS AND REFERENCES

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13.12 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) A derivative is a financial instrument that derives its value from an underlying asset, such as a stock, bond, currency, or commodity. The value of a derivative is based on the changes in the value of the underlying asset.

2) Derives its value from an underlying asset, traded globally, Exchange-traded derivatives like futures or stock options are standardized and eliminate or reduce many of the risks of over-the-counter derivatives, usually leveraged instruments.

3) Lock in prices, hedge against risk, can be leveraged, useful in making a diversified portfolio. Limitations: difficult to value them, subject to counterparty default, complex, sensitive to demand and supply factors.

Check Your Progress 2

1) An option is an agreement between two parties (say A and B) in which a party, say, A, has the right to buy or sell a particular asset at a predetermined price within a specified time frame.

2) A **call option** is a type of derivative in which the buyer gets a right to **buy** an underlying at an agreed price. A **put option** is a type of derivative in which the buyer gets a right to sell an underlying at an agreed price.

3) No, Futures are a contract that obligates the buyer to purchase or sell an asset at a specified future date and price, while options give the buyer the right, but not the obligation, to purchase or sell an asset at a specified price and date.

4) Refer to Table 13.2 and 13.3

Check Your Progress 3

1) Refer to Section 13.8

2) Refer to Section 13.8